

✓ PHASE-WISE CHECKLISTS (2-Week Blocks)

PHASE 1 (Weeks 1–2) — Hardware Planning & Ordering

Checklist:

- ☐ Create preliminary BOM (Bill of Materials)
 - ☐ Confirm microcontroller choice (e.g., Teensy 4.0)
 - ☐ Confirm motor/driver compatibility
 - ☐ Confirm camera model and interface
 - ☐ Verify PSU voltage/current requirements
 - ☐ Identify suppliers inside Germany/EU
 - ☐ Place all orders and track shipping
 - ☐ Create organizing folder for datasheets
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PHASE 2 (Weeks 3–4) — Beginner Foundations

Checklist:

- ☐ Install Teensy/Arduino IDE + toolchain
 - ☐ Blink LED on Teensy (GPIO test)
 - ☐ Generate PWM on a pin
 - ☐ Read Stepper driver datasheet
 - ☐ Learn basic PID concept (P, I, D roles)
 - ☐ Run simple PID simulation (optional MATLAB/Python)
 - ☐ Document learning progress briefly
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PHASE 3 (Weeks 5–6) — Initial Hardware Testing

Checklist:

- ☐ Wire Teensy to one stepper driver
- ☐ Connect driver to stepper motor
- ☐ Verify PSU output voltage
- ☐ Test step pulses + direction signals

- ☒ Move motor a fixed number of steps
 - ☐ Log behaviour (speed, noise, vibration)
 - ☐ Create safety notes (max current values)
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PHASE 4 (Weeks 7–8) — Single-Actuator Control

Checklist:

- ☐ Implement simple PID for position control
 - ☐ Tune P term to get basic response
 - ☐ Add I+D terms once stable
 - ☐ Mount simple test platform
 - ☐ Investigate camera model options (fps/latency)
 - ☐ Take notes on actuator responsiveness
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PHASE 5 (Weeks 9–10) — Mechanical Platform Design

Checklist:

- ☒ Install Fusion 360 or SolidWorks
 - ☒ Model arm & motor mounts
 - ☒ Model platform geometry
 - ☒ Export CAD to STL/DXF if needed
 - ☒ Review torque requirements & arm lengths
 - ☒ Validate assembly fits together
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PHASE 6 (Weeks 11–12) — Manufacturing & Assembly

Checklist:

- ☒ 3D print/CNC all mechanical parts
- ☒ Sand/finish fits if required
- ☒ Assemble frame + platform
- ☒ Install motors + mounts
- ☒ Wire power, ground, signal neatly

- ☒ Add cable management (zip ties, sleeves)
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PHASE 7 (Weeks 13–14) — Multi-Actuator Bring-Up

Checklist:

- ☒ Connect 2–4 stepper drivers to Teensy
 - ☒ Move motors in coordinated pattern (sin wave/sequence)
 - ☒ Test PSU load under motion
 - ☒ Log actuator temperatures (optional)
 - ☐ Check for mechanical interference
 - ☐ Adjust step rates for smoothness
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PHASE 8 (Weeks 15–16) — Camera Integration

Checklist:

- ☒ Test camera feed in Python (OpenCV)
- ☐ Measure real FPS & resolution
- ☒ Implement ball detection (color or circle detection)
- ☐ Display ball position overlay (visual debug)
- ☐ Log (x, y) ball coordinates
- ☐ Document lighting requirements

End of

PHASE 9 (Weeks 17–18) — Vision + Control Integration

Checklist:

- ☐ Map camera pixels → platform coordinates
 - ☐ Send ball coordinates to Teensy (USB/Serial)
 - ☐ Implement closed-loop control logic
 - ☐ Adjust delays & sampling rates
 - ☐ Validate feedback with slow bounces
 - ☐ Record short test videos
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PHASE 10 (Weeks 19–20) — Full Closed-Loop Control

Checklist:

- ☐ Tune PID for multi-axis control
 - ☐ Achieve continuous bounce
 - ☐ Test different ball drop heights
 - ☐ Measure latency end-to-end
 - ☐ Video record stable operation
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PHASE 11 (Weeks 21–22) — Experimental Evaluation

Checklist:

- ☐ Define performance metrics (e.g., stability, tracking error)
 - ☐ Run repeated trials
 - ☐ Save raw data logs
 - ☐ Plot ball trajectory vs platform motion
 - ☐ Summarize results in a lab notebook
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PHASE 12 (Weeks 23–24) — Analysis & Improvements

Checklist:

- ☐ Identify control instability causes
 - ☐ Improve mechanical stiffness if needed
 - ☐ Improve PID or filtering (e.g., Kalman optional)
 - ☐ Re-run experiments post improvements
 - ☐ Capture final demo videos
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PHASE 13 (Weeks 25–26) — Documentation Part 1

Checklist:

- ☐ Write Introduction + Problem Statement
- ☐ Write Background + Related Work
- ☐ Write Hardware & Mechanical Design

June - July

- ☐ Create system block diagrams
 - ☐ Insert CAD screenshots + BOM
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PHASE 14 (Weeks 27–28) — Documentation Part 2

Checklist:

- ☐ Write Control & Vision Methods
 - ☐ Write Experiments + Results
 - ☐ Write Discussion + Limitations + Future Work
 - ☐ Format references + appendices
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PHASE 15 (Weeks 29–30) — Revision & Submission

Checklist:

- ☐ Proofread entire thesis
- ☐ Check formatting rules (university style)
- ☐ Generate PDF + backups
- ☐ Prepare defense slides
- ☐ Submit thesis
- ☐ Demo prototype (optional)

WEEKLY STUDY RESOURCE LIST (Beginner-Friendly)

Below are structured resources aligned to phases:

Weeks 1–2: BOM & Hardware

Study:

- Basic component datasheets
- How to choose steppers + drivers
- Basic voltage/current calculations

Resources:

- YouTube: “Stepper motors explained”
- Article: “How to choose a power supply”
- Datasheets from OMC/StepperOnline

Weeks 3–4: Microcontroller + PID Basics

Study:

- Teensy/Arduino programming
- PWM + Serial debugging
- PID basics

Resources:

- “PJRC Teensy getting started”
- Book: *PID Without a PhD* (Simple)
- YouTube: “PID control intro for beginners”

Weeks 5–6: Stepper Control

Study:

- Step/dir motor driving
- Microstepping
- Current limiting

Resources:

- “How stepper drivers work” videos
- OMC StepperOnline driver guides
- Arduino Stepper library examples

Weeks 7–8: Single Axis Control + Camera Basics

Study:

- PID tuning strategies
- Camera exposure & FPS concepts

Resources:

- MATLAB PID tuning demo (optional)
- YouTube: “Camera shutter, exposure, FPS basics”

Weeks 9–10: CAD Modelling**Study:**

- Basic 3D modelling
- Joint mounting + tolerances

Resources:

- Fusion 360 “Designing Enclosures”
- YouTube: “Parametric modeling basics”

Weeks 11–12: Mechanical Assembly**Study:**

- CNC/3D printing tolerances
- Cable management
- Power wiring safety

Resources:

- 3D printer calibration guides
- EEVblog power safety basics

Weeks 13–14: Multi Actuator Control**Study:**

- Coordinated motion control
- Power distribution concepts

Resources:

- YouTube: “Coordinated motion control basics”
- Teensy Forum multi-axis projects

Weeks 15–16: Computer Vision (OpenCV)

Study:

- Color tracking
- Shape detection (Hough Circles)

Resources:

- OpenCV-Python Docs (ball tracking examples)
- YouTube: “OpenCV detect ball tutorial”

Weeks 17–18: Vision + Control Integration

Study:

- Serial communication PC → MCU
- Coordinate transformation

Resources:

- Tutorials on Serial/USB communication
- OpenCV + microcontroller integration blogs

Weeks 19–20: Real-Time Tuning

Study:

- Latency measurement
- Control loop frequency

Resources:

- YouTube: “Real-time control basics”
- Embedded timing tutorials

Weeks 21–22: Experimental Design

Study:

- How to collect data
- How to analyze response curves

Resources:

- SciPy/NumPy plotting tutorials
- MATLAB alternative: Python Matplotlib

Weeks 23–24: System Optimization

Study:

- Signal filtering (Simple low-pass)

- Mechanical damping concepts

Resources:

- YouTube: “Low-pass filter basics”
- Blogs on mechanical vibration reduction

Weeks 25–30: Documentation

Study:

- Thesis formatting
- Graph creation (Python/MATLAB)
- Technical writing clarity

Resources:

- “How to write a scientific thesis”
- Overleaf or Word templates
- IEEE citation basics